Magnetide v21-r2i — Design Document

Reynard Filbert Tang · Kelas 10.8, Kelompok 1 · MHD seawater pump · revision r2i · all dimensions mm, tolerances ±0.2 unless stated

1. System summary

Function	Conduction MHD pump — Lorentz force on seawater in a crossed field
Envelope	150 × 81 × 45
Bore	40 × 14 × 124 (W × H × L)
Mass, steel	2 × 376 g
Mass, plastic	≈ 150 g (4 printed parts + 2 caps)
Field in bore	483 mT design, N52, 96 tiles
Closure	8 × M3 through cropped lugs, steel-to-steel at the split
Sealing	Tray is a closed cavity; no gasket at the split (see §7)

2. Part list

Part	Qty	Material	Size (X×Y×Z)	Mass	Process
Yoke half	2	3.0 mm mild steel DC01/SPCC	150 × 81 × 22.5	376 g	laser + 4 bends
Tray	2	PLA or PETG	124 × 50.7 × 19.5	≈ 30 g	FDM
Cap	2	PLA, 100 % infill	121.7 × 42.7 × 1.0	≈ 5 g	FDM
Bellmouth	1	PLA or PETG	20 × 52 × 39	≈ 21 g	FDM
Rear block	1	PLA or PETG	5.5 × 52 × 39	≈ 8 g	FDM
Electrode	2	graphite plate	134 × 14 × 2	≈ 16 g	bought / cut
Magnet tile	96	N52 NdFeB, Ni-Cu-Ni	20 × 10 × 5	≈ 7.5 g	bought
Bolt set	8	A2 stainless	M3 × 14 + nut + washer	—	bought
Lead grommet	2	epoxy	Ø1.6	—	field-applied
Total mass, dry					**≈ 1.15 kg**

3. Steel yoke

Parameter	Value	Note
Sheet thickness	3.0	one gauge, plate and wall (+0.4 kg total)
Blank	106.43 × 150	stepped outline — lugs only, not a band. Bend development, not an outside-dimension sum
Inside bend radius	3.0 / 1.5	R3 wall folds, R1.5 lug folds — an 8 mm lug would leave only 3.5 mm of flat — too little for a Ø3.4 hole
K-factor	0.44	mild steel at R/t = 1.0 (R3) and 0.5 (R1.5)
Bend allowance	6.786 / 4.430	(R + K·t)·π/2, wall / lug
Bend 1 — wall	at ±25.89	90° down, full length; arc centre on the blank
Bend 2 — lug	at ±43.50	90° out, two lugs per side at x 40 / 110

Lug size	24 × 12	two Ø3.4 holes per lug, 13 pitch, hole line 6.0 off the plate face. 12 mm is the minimum: the flat flange is 7.5 mm and a Ø3.4 hole needs 5.4 plus margin
Flux path	—	plate → wall → lug face → facing lug → wall → plate

> The lug is cropped rather than continuous because the old yoke looked better that way, and the clamp load is identical: 8 × M3 at 2 N·m gives ≈ 8 kN total preload across four lug pairs. What the crop does cost is torsional stiffness at the split — the lugs locate on bolt shanks alone, so add two 3 mm dowels if the pump gets knocked around.

4. Tray — the water barrier

Feature	Value	Why
Floor thickness	1.5	only material between bore and magnets
Cavity	120 × 40 × 10	48 tiles per face, 2 layers of 5 mm
Wall at bore	4.85	seals the bore side and backs the electrode
Electrode groove	1.0 deep × 2.2 wide	ceiling sits 0.5 *below* the cavity floor
Peg holes	Ø1.65 × 1.6 blind	0.4 mm of plastic left in a 2.0 wall
Lead channel	Ø1.6 blind, 0.4 skin	drill through at wiring, seal with epoxy
Tray-to-yoke clearance	0.15 / side	drop-in; see §7 on shimming
Cap ledge	1.0 × 1.5	lid sits flush with the steel inner face

5. Magnetic and electrical design

Parameter	Value	Basis
Magnet grade	N52	Br ≈ 1.43 T
Tiles per face	48	6 × 4 × 2 layers of 5 mm
Active pole area	120 × 40	4800 mm^2
Field in bore	483 mT	2 mm effective gap (1.0 cap + steel), measured on v21
Electrode area	124 × 14 = 1736	mm^2 wetted, per plate
Electrode gap	40	bore width
Current density, design	0.1 A/cm^2	graphite in seawater, conservative
Design current	1.7 A	0.1 A/cm^2 × 17.4 cm^2
Lorentz pressure	≈ 82 Pa	J × B × gap, idealised
Blocked head	≈ 8 mm H_2O	p / (ρg), seawater 1025 kg/m^3

> Treat these as an order-of-magnitude check, not a performance guarantee. Real throughput will be lower: shunt current through the seawater beside the plates, end losses at the inlet and outlet, and bubble void fraction from electrolysis all reduce the ideal numbers. The v21 boat test measured 0.2–0.3 m/s with a straight outlet — that is the number to design the hull around.

5.1 Why the flux path matters more than the magnet grade

Every millimetre of non-magnetic material between magnet and steel is in series with the whole circuit. The cap is 1.0 mm and the steel plate is directly above it, so the gap is 1.0 mm of plastic. Doubling that cap to 2 mm would drop the bore field by roughly a fifth. This is also why nothing compressible goes on the cap face — foam there is an air gap wearing a plastic hat.

6. Fit and tolerance table

Interface	Nominal	Clearance	Failure mode
magnet ↔ cavity	9.7 in 10.0	0.3 total	press fit cracks tiles
cap ↔ ledge	121.7 in 124	0.3 total	cap lifts, stack moves up
tray ↔ yoke throat	50.7 in 51.0	0.15 / side	rattle, floor wear
electrode ↔ groove	2.0 in 2.2	0.2	water bypasses the plate
peg ↔ hole	3.0 in 3.3	0.3	end block misaligns
M3 ↔ lug hole	3.0 in 3.4	0.4	standard clearance
split faces	0	metal-metal	any shim = field loss

7. How the yoke retains the tray

Vertically the assembly is a stack: tray rim → steel plate (z 19.5), mirrored tray butted at z 0. Bolting the halves closes that stack, so the plastic cannot move up or down once the lugs are tight. Lengthwise, the bellmouth and rear block pin the tray at both ends through the pegs and electrode pockets. Sideways there is **0.15 mm a side of clearance and no clamp** — the bolts never squeeze the plastic, they only pull lug to lug.

> If the tray rattles sideways, a 0.3–0.5 mm closed-cell foam strip on the outer wall faces will take it up. Do not put foam between cap and steel plate: that surface is the magnet's path into the yoke. Do not use open-cell foam anywhere in the assembly — it wicks brine into the cavity.

8. Revision history

Rev	Change	Reason
r2c	cap introduced, per-tile magnets	keep magnet stack captive
r2e	magnet moved inside a closed tray cavity	tray should enclose, not support
r2f	groove 3.0 → 1.0 deep; gasket added	groove roof sat 1.5 mm above the cavity floor
r2g	flange cropped into lugs; peg holes blind; lead exit blind	restore the old yoke look; close through-holes
r2h	lugs 20 → 24 mm with **two holes each**	two fasteners per lug, per the reference sketch
r2i	sheet 2.5 → 3.0 mm; R0.2 → R3 wall / R1.5 lug; blank 107.83 → 106.43; lug 8 → 12	R0.2 on 2.5 mm is 86 % outer-fibre strain — it cracks; 3 mm buys ~17 % more gap field, 4 mm is the model peak

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